Long-Term and Extensive Monitoring for Bee Colonies Based on Internet of Things

Wei Hong, Baohua Xu, Xuepeng Chi, Xuepei Cui, Yinfa Yan, and Tongyang Li

***Abstract*—The pollination during bees' foraging is vital to continue species on the Earth. However, the losses of bee colonies are being more extensive problems in recent decades, which have seriously impacted the global environment but have not received enough attention from human beings yet. Based on widely investigating detectable features, this article presents an IoT-based system for long-term and extensive monitoring of bee colonies. Multiple features, including temperature and humidity inside the hive, bee combs' weight, bee colony's sounds, and a number of bees passing through the hive entrance, can be detected hourly by the proposed system. Furthermore, a four-month practical monitoring indicates that the system can continuously operate without human intervention and the data can reveal the activity and growth of bee colonies. Especially, a swarming activity was successfully observed by the data, which shows a significant potential to recognize various activities of bee colonies. The highlights of this system are the capabilities of detecting diverse features, transmitting data wirelessly and that with a strong self-support ability. Because of these advantages, the system can operate for an exceeding long time without maintenance, which will largely benefit system deployment in the field. In the visible future, this system could contribute to the study of bee's behavior and the development of precision beekeeping.**

***Index Terms*—Bee colony monitoring, Internet of Things, precision beekeeping, smart hive.**

I. Introduction

HONEY bees are important social insects that can produce many valuable and useful products, such as honey and royal jelly. Meanwhile, they pollinate about 35% of the food crops in the world [1]. However, the losses of bee colonies are being more extensive problems in the recent decades [2]–[7], which caused by many implicated factors, such as honey bee parasites, pathogens, use of pesticide and antibiotics, climate change, nutritional stress, and so on. The study according to the National Agricultural Statistics Service [8] mentioned that the number of managed honey bee colonies used for honey production in the USA has decreased steadily since the late 1940s. Moreover, a national survey which involved 5937 valid participants and 414 267 managed colonies [9] reported that the average loss in the USA was 49% in 2014–2015, and it also reviewed the history data that the loss rates were 36% in 2007–2008, 22% in 2011–2012, 45% in 2012–2013, and 34% in 2013–2014. These phenomena imply some serious problems in the bee's ecosystem, which would significantly impact the agricultural industry and global environment, but have not received enough attention from human beings yet.

In order to understand the activities of the bee colony, Gates [10] began to monitor the bee colony's temperature as early as 1907. In the following centuries, many more quantified techniques were applied to monitor bee colony [11], e.g., by using weight [12], [13], temperature [14]–[18], humidity [19]–[21], gas concentration [22]–[24], vibration [25]–[28], sound [29]–[33], and entrance counts [34]–[40]. These studies showed a close relationship between the detected features and bee colony's activities, which established the foundation of continuous monitoring for bee colonies. With the development of electronic information technology, the precision beekeeping based on continuous monitoring of individual bee colonies to minimize resource consumption and maximize the productivity has become a hotspot in the past few years [41]–[44]. However, available monitoring systems for bee colonies are still absent due to some engineering challenges [45]. First, the bee colony is a complex system with thousands of individuals in which a single sensor is not able to fully recognize the various patterns of bee colonies. So how to combine multiple sensors within a monitoring system as much as possible is a crucial problem to improve the cognition accuracy. Second, wired data transmissions would cause a lot of troubles because bee hives may be moved frequently in the actual beekeeping, so how to transfer the data in a wireless way is also essential. Third, the capacity of power self-support determines whether the system can operate continuously in the field.

The vision of IoT is to build a smart world by sensing everything which has been espoused over the world [46]. With the feasibility of using the Internet almost everywhere, IoT's principle, framework, and technology are being more available [47], which bring a big opportunity to develop smart agricultures [48], [49], especially for beekeeping which has high requirements for timely knowing the status of each colony

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and frequently changing the strategy based on environment conditions [50]. Based on widely investigating the detectable features of bee colonies, this article presents an IoT-based system with multiple sensors, wireless data transmission, and power self-support for long-time and extensively monitoring of bee colonies, which is outstanding in the capabilities of detecting diverse features, transmitting data wirelessly and that with a strong self-support ability, that makes the system more practice and valuable for the study of bee's behavior and the development of precision beekeeping. The remainder of this article is organized as follows. Section II investigates several detectable features of bee colonies. Section III describes the design of the monitoring system. Section IV provides the analysis of monitoring results. In Section V, some conclusions are drawn.

II. Investigation of Detectable Features

A. Weight

The weight of the bee hive is a common indicator to evaluate bees' quantity, stock of honey, and harvesting time. More abundant information about bee colony's activities can be obtained from continuous weight monitoring with a satisfied resolution. Hambleton [12] hourly weighed a bee hive in several days by manual and found that the weight change has a strong characteristic that the weight decreases during midnight and early morning because of nectar's consumption and bees' leaving, and the weight increases in the daytime because of foraging activity. Meikle *et al.* [13] applied an electronic balancer to four bee colonies for about 17 months, and the results indicate that the weekly change of the average weight of the bee hive can represent the change of stored food, and the daily amplitude of the detrended weight can represent the daily consumption of food. Therefore, these two indices of weight data can reveal the colony's activity and growth without disturbing bees. Moreover, the weight change caused by swarming was detected in their study, which could be evidence to identify swarming activity.

B. Temperature

Honey bees are poikilotherms so that they need to cluster together to make their body temperature above 15 °C and keep their larvae around 35 °C for growing [14]. So, the temperature maintenance is a crucial work for a bee colony: bees will cluster together and generate metabolic heat to warm up their bodies when the ambient temperature is too low, and they will increase the humidity through bringing back water from outside and cool down the colony through fanning when the ambient temperature is too high [16]. Therefore, the temperature stability of the bee colony can reflect the colony's adaptive capacity and healthy level. Kridi *et al.* [17] installed a temperature sensor in the center of the bee's comb to diagnose the colony's status according to temperature stability. As bees need to warm up their bodies before flight [15], there is an evident rising of the colony's temperature in swarming. Zacepins *et al.* [18] reported that the temperature of the upper hive raised from 34 °C–35 °C to 37 °C–38 °C, which is proposed as a signature of swarming.

C. Humidity

The humidity in the hive is very important to the growth of larvae [21]. Doull [19] studied the hatching of bee eggs under different humidity, and he found that the optimal humidity is 90%–95%RH and the eggs cannot be hatched when the humidity is below 50%RH. However, the humidity in the hive is susceptible, which is not only passively affected by environmental factors, such as ambient humidity, temperature in hive, and nectar's moisture, but also actively adjusted by bee colony's activities, such as carrying water, feeding larvae, and isolating behaviors. Human *et al.* [20] used several temperatures and humidity sensors to simultaneously monitor the ambient condition, breeding comb, and nectar areas, and the result indicates that the humidity of breeding comb is the highest (about 40%RH) and has the best stability. Obviously, the monitored humidity is quite lower than the optimal humidity mentioned by Doull, which further indicates that the humidity control under extreme weather is a serious problem for bee colonies.

D. Gas Concentration

O_2 and CO_2 are almost involving in all the metabolisms of organisms on Earth. Seeley, respectively, introduced the pure gases of O_2, CO_2, and N_2 into a bee hive and found that the quantity of fanning bees is proportional to CO_2's concentration. They further speculated that bees are seriously controlling the concentration of CO_2, because CO_2 can significantly influence the function of succinic dehydrogenase which is essential to redox reaction [22]. Actually, the concentrations of CO_2 and O_2 in the hive are changing periodically as animal breathing, where the amplitudes are about 0.6% and the frequencies in the daytime are around 2.9 times per minute and about seven times higher than that at night [23]. Besides, the bee colony can actively maintain a low concentration of O_2 to slow down their metabolisms to survive from the shortage of food in winter [24].

E. Vibration

Vibration plays a crucial role in bee colony's information spreading and action coordinating [26]. Michelsen *et al.* [25] applied electromagnetic force to bee comb to simulate two typical vibrations generated by the queen bee before swarming and then studied their effects on the bee colony. Nieh and Tautz [27] used a laser vibrometer to measure the vibrations of bee's waggle and analyzed the signals' characteristics through the Fourier transform, which verified that the receptor on bees' leg can detect the vibrations in the range of 200–300 Hz. Bencsik *et al.* [28] used an accelerator to further study hive's vibrations from 0 to 4000 Hz and they found that the amplitudes on 500 and 2000 Hz can hint the swarming activity.

F. Sound

The sounds in a bee hive are like the vibrations where different frequencies are related to different colony activities. As the sounds can be heard by humans, their meanings have been revealed more than that of vibrations in the past. As early as 1957, Frings and Little [29] studied bees' response to sound waves with different frequencies and they found that almost all the bees stop moving when the sound waves, including frequencies of 300–1000 Hz, are playing with sufficient intensity. Esch [30] focused on the sounds produced during swarming, and he found that the bees joined in swarming would buzz their wings every 0.5–3 s with frequencies of 180–250 Hz and produced a permanent noise with frequencies of about 500 Hz when they were contacting a hive mate. Ishay and Sadeh [31] studied the sounds of Apis mellifera ligustica, Apis cerana, and Vespa manderinia in some specific activities such as ventilating, and the result indicated that the bees' purposes could be known through the frequencies of their sounds. Furthermore, some researches [32], [33] also verified that bee colony's sounds could be used to identify and prevent the occurrence of swarming.

G. Entrance Counts

In a bee colony, most of the bees above 18 days old are responsible for foraging, so the capability of foraging can be known through counting the bees' number of leaving and returning hive. Some manual [35], [36] and automatic [34], [37]–[40] methods for obtaining the number were presented. Corbet *et al.* [36] manually recorded the leaving number with the hive's temperature, and they found that there is an evident threshold of temperature for bees' leaving and the leaving number is proportional to the temperature in a certain range. In order to study the colony's activities in normal, poisoning, and swarming, Struye *et al.* [37] used a photoelectrical entrance counter which can distinguish the moving directions of bees and the result showed the significant values of the entrance counter. The same device was also applied to study the flight activity of bees under several conditions, such as with different colony size, ambient temperature, and the time of day [40]. Moreover, radio-frequency identification (RFID), as a novel technique for tracking specified objectives, can be used to monitor individual activities through adhering a micro metal label onto the bee's back [38], [39].

III. Design of Monitoring System

As the bee is a social insect that has various behaviors, using only one feature may not be able to identify all the activities of bees. Therefore, combining the detectable features as many as possible can greatly benefit the activity identification of the bee colony. However, for a practical design, other factors, such as the relevance to bee colony's activities, the richness of information, the cheapness of cost, and the simplicity of engineering, should also be considered. In this article, the four considered factors are quantified between the above detectable features. For example, the weight is strongly related to the colony's activities, such as foraging and honey storage, so the score of "relevance" is high; the sensor resolution of weight is better than that of humidity but the data may not provide so much information, such as vibrations and sounds of the colony, so the score of "richness" is medium; as the weight sensor is more expensive than others, the score of "cheapness" is low, and since some machining is required for the installation of a weight sensor, the score of "simplicity" is low. The other detectable features are evaluated in the same way as shown in Table I.

TABLE I
Applicability Evaluation for General Detectable Features

Detectable Features	Relevance	Richness	Cheapness	Simplicity
Weight	High	Medium	Low	Low
Temperature	High	Medium	High	High
Humidity	Low	Low	High	High
Gas Concentration	Low	Medium	Low	Medium
Vibration	Medium	High	Medium	High
Sound	High	High	High	High

Since the relevance is recognized as the highest priority in this article, four detectable features (weight, temperature, sound, and entrance counts) are considered in the developed system, which are the same as Zacepins' suggestion [45]. Moreover, humidity is also included because it is commonly integrated into a temperature sensor and no more cost needs to be paid.

The sensors' location will seriously affect the physical meaning and the signal-to-noise ratio of the detected data. The weight sensor is usually designed in the bottom of the bee hive to measure the total weight, which would be influenced by ambient temperature, humidity, and blowing wind [10], [12]. In this article, a weight sensor is installed in the hive to measure the weight of bee combs only. Outside the hive, inside the hive, and on bee comb are three typical locations for temperature and humidity sensors. Apparently, installing the temperature and humidity sensor outside the hive is difficult to obtain the bee colony's status. Although the data from bee comb are strongly related to the current status, it lacks diagnosing capacity for the bee colony's health because the data are usually kept stable until some serious problems occur [17], [20]. Thus, installing temperature and humidity sensor inside the hive is better than the other two strategies. As sounds can freely spread in a bee hive, acoustic sensors can be directly installed in the hive. In order to know the total number of bees exiting and entering the bee hive, the entrance sensor needs to cover all the channels of the bee hive.

Fig. 1 presents the framework of the whole monitoring system, where the smart hive is a common bee hive combined with a detection device which can automatically perform some tasks such as information collections through the onboard electronic components, the router is a commercial product for creating a Wi-Fi hotspot for the smart hives from the mobile network, the mobile base stations are available infrastructures of telecom carriers, and the cloud server is a virtual computer which comes from IT service providers, such as Amazon and Alibaba. In this system, each smart hive continuously collects bee colony's information and the information is wirelessly relayed by the cloud server's database through the mobile Wi-Fi routers and available mobile base stations. The cloud server then takes responsible for storing, processing, and displaying the information. Meanwhile, the commands coming from users or embedded expert strategies will be returned

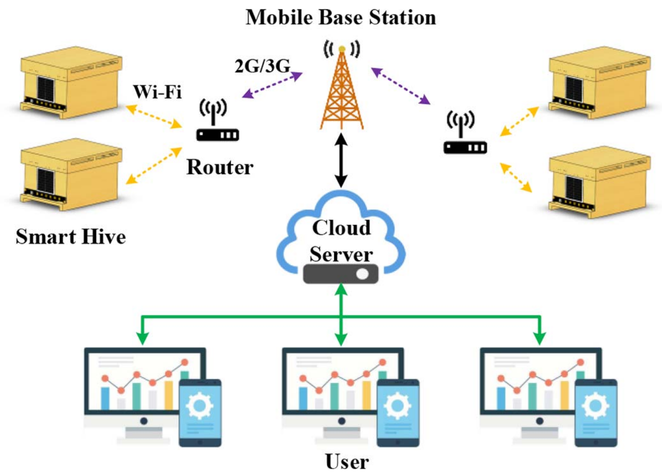

Fig. 1. Framework of the whole monitoring system.

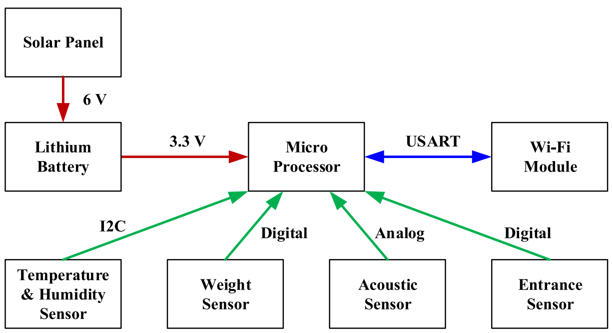

Fig. 2. Block diagram of the detection device.

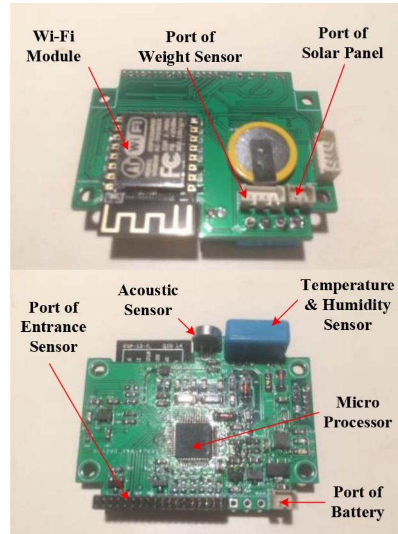

Fig. 3. Photograph of main board of the detection device.

TABLE II
AVERAGE CURRENT ESTIMATION OF SMART HIVE

Item	Model	Average Current
Micro Processor	STM32	75 uA
Wi-Fi Module	ESP8266	583 uA
Temperature & Humidity Sensor	DTH12	1 uA
Weight Sensor	YZC-133	1 uA
Acoustic Sensor	4.5*2.2	1 uA
Entrance Sensor	CAP	58 uA
Total		719 uA

TABLE III
PERFORMANCE OF SMART HIVE

Item	Paramter
Hive Size	37cm×47cm×26cm (Inside)
Entrance Size	48mm×7mm
Power Supply	Lithium Battery+ Solar Power
Data Transmission	WiFi
Tepmature Detection	-20~60℃±0.5℃
Humidity Detection	20~95%RH±5%RH
Weight Detection	0~20Kg±2%
Acoustic Detection	>-40dB
Entrance Detection	0~10000/h

to the detection device through the same paths for setting working modes and system parameters, such as sample rate, local time, and firmware update. Finally, the users can get the bee colonies' statuses conveniently and manage their apiaries on line.

Since the cloud server and mobile base stations are available infrastructures and the mobile Wi-Fi routers are mature products, the hardware design is mainly for the detection device installed in each smart hive. Fig. 2 is the schematic of the detection device. The solar panel is used to convert solar energy to electricity for maintaining the operation of the device, and the lithium battery is used to store energy and make a power balance between generation and consumption for a short time. The Wi-Fi module is configured to communicate with the cloud server. The four sensors are used to obtain the above-mentioned features of bee colonies. The microprocessor takes responsible for coordinating the operation of all the components. The mainboard of the detection device is shown in Fig. 3.

In order to achieve power self-support, it is necessary to reduce device power consumption as much as possible. Except for using low-power components, an intermittent working mode is also applied in this design to further control the consumption. Meanwhile, the solar panel and battery with enough capacity are required to cover the annual consumption and support the device's operation during the time when there is no sunlight. In this design, the device collects and transmits the data once an hour, and the average current of the device is estimated in Table II. In the practical testing, a 400-mAh battery can support the device operating for more than two weeks in a complete dark environment, and the battery can almost maintain full of energy when a 2-W solar panel is plugged on the device.

For the convenience of device production and system deployment, most electronic components are designed in a main module which is installed across the wall of bee hive for the sensors to detect the internal information as shown in Fig. 4. The performance of the smart hive is shown in Table III.

Since the developed system is using the commercial Wi-Fi module and router which is based on the mature network protocols, such as IEEE 802.11 and TCP/IP, the data transmission can be guaranteed. Therefore, the design of software mainly

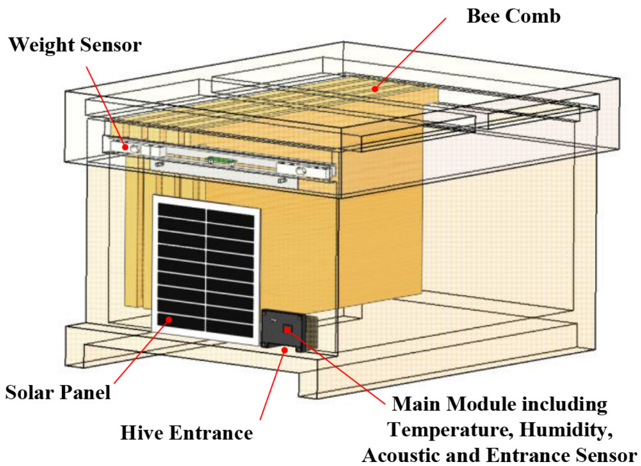

Fig. 4. Structure of smart hive.

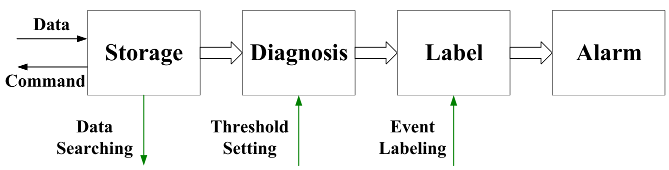

Fig. 5. Application architecture of the cloud server.

Farm	ID	Time	Power	TempL	TempR	HumiL	HumiR	CountL	CountR	Door	Cap	Weight	Night	Day
1	1:7799	2018-12-06 16:00	3.65	13.5	15.3	14.7	10	0	0	1	0	2508	106	163
1	2:1502	2019-12-04 11:00	4.11	15.4	0	10	0	12	0	1	0	342	-381	-285
1	3:9213	2019-12-04 11:00	3.99	12.3	12.2	95	15.5	0	0	1	0	-220	-2	-7
1	4:6048	2019-12-04 11:00	3.93	11.7	10.7	10	16.6	0	0	1	0	113	-3	0
1	5:7604	2019-12-04 11:00	3.82	12.5	11.2	10	32.5	0	0	1	0	332	0	0
1	6:9469	2019-12-04 11:00	4.02	0	11.4	0	95	0	0	1	0	-106	-16	-19
1	7:1485	2019-12-04 11:00	4.08	12.5	11.6	10	52	0	0	1	0	-80	-1	1
1	8:7516	2019-12-04 11:00	4.17	45.5	45.5	95	95	4993	4417	1	0	4893	-239	416
1	9:2940	2019-12-04 11:00	4.17	17.7	17.7	10	10	0	0	1	0	-537	0	0
1	10:2674	2019-12-04 11:00	4.07	13.3	13.3	22.2	22.2	0	0	1	10:29	-7734	-99	-280
2	1:1095	2018-10-15 10:00	3.8	20.4	20.4	70.5	70.5	0	0	1	0	46	12	12
2	2:6775	2019-09-02 00:00	4.14	25.5	25.5	90.4	90.4	0	0	1	0	-535	22	22
2	3:3348	2019-08-17 08:00	4.19	26.5	26.5	13.2	13.2	0	0	1	0	-37	-7	76
2	4:7895	2018-10-15 10:00	3.82	20.6	20.6	70.1	70.1	0	0	1	0	-43	-1	-1
3	1:1554	2019-12-04 11:00	4.02	14.1	13.1	12.7	11.6	0	0	1	0	0	0	0
3	2:9785	2019-12-03 09:00	3.88	20.1	19	10	10	0	0	1	0	31	2	2
3	3:5339	2019-09-26 15:00	4.06	28.7	27.8	28.5	38.5	4	891	1	0	0	-438	-438
3	4:9786	2019-09-27 10:00	4.15	25	25	93.2	89.8	9	11	1	0	547	0	0
4	1:8312	2019-12-04 11:00	4.3	8.9	11.3	37	20.7	0	1	1	0	3203	-320	165
4	1:1048	2019-10-24 18:00	3.72	19.1	18.7	67.7	64.4	0	0	1	0	3247	12	2
4	2:9559	2019-12-02 06:00	3.75	13.2	13	59.5	52.5	0	0	1	0	197845	0	0
4	2:6553	2019-10-17 13:00	3.76	15.6	14.2	64.2	65	0	0	1	0	3220	-37	2
4	3:0568	2019-12-04 11:00	4.21	10.8	11.3	93	36.1	0	12	1	0	4202	-248	-248
4	3:1917	2019-10-23 20:00	3.73	16.5	17.3	74.1	70.4	0	0	0	0	3409	9	-9
4	4:0253	2019-12-04 11:00	4.27	10.9	11.1	23.1	35	0	50	1	0	2030	-14	-9
4	4:9783	2019-10-15 05:00	3.76	7.9	8.3	55.1	55.5	0	0	1	0	3717	-18	-18

Fig. 6. Data display through a website.

focuses on the layer of an application. The application of the cloud server is realized through PHP + MySQL and the architecture is shown in Fig. 5. After transmissions by Internet, the data of the bee colonies are stored in the database (MySQL), and then the specified command for each smart hive is returned by the same paths. Based on the updated data, the application will diagnose the status of bee colonies through the thresholds came from the manual setting, and the data will be labeled with the diagnosis result and the manual input for the offline analysis and improving the diagnosis. Furthermore, the abnormal status will be alarmed to beekeepers by mobile instant messages. Meanwhile, the user can browse the data through a website as is shown in Fig. 6.

IV. Analysis of Monitoring Result

Ten smart hives are manufactured and placed in E110.81°N24.85° for monitoring several colonies of Apis cerana, as is shown in Fig. 7. The cost of each smart hive is about \$200, which can be reduced by mass production.

Fig. 7. Photograph of system deployment.

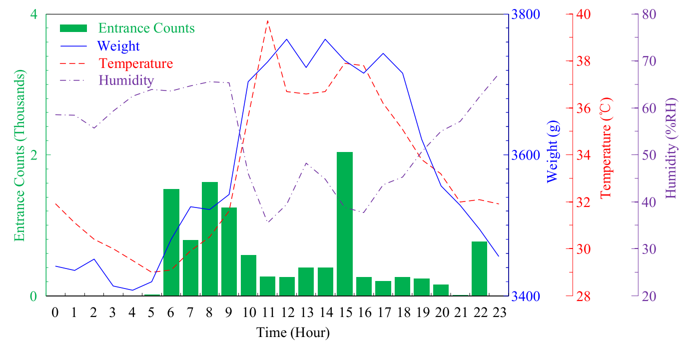

Fig. 8. Daily monitoring data.

A. Routine Activity Observation

Fig. 8 provides a set of daily monitoring data of the bee colony numbered 9469 on June 19, 2018, where the green bar represents the hourly counts of bees passing through the hive entrance, the blue solid line represents the hourly weight of all bee combs, the red dashed line represents the hourly temperature inside the bee hive, and the purple dashed-dotted line represents the hourly humidity inside the bee hive. The result indicated that the foragers largely went out for food after 5:00 so that the combs' weight was increasing with the time, the foraging went down after 9:00 because the inside temperature went so high that may compel the foraging bees to cool down the colony, a lot of entrance counts appeared during 14:00–15:00 which is caused by practice flights of the newborn bees, the inside temperature went down and the humidity went high after 16:00, the weight apparently went down after 18:00 because the foraging was stopped but the feeding was still going on, and there was an unknown event at 22:00 which caused a peak in the entrance counts.

Fig. 9 presents the 120-day monitoring data of the bee colony numbered 9469 from February 18, 2018 to June 19, 2018, where the green bar represents the daily entrance counts, the blue solid line represents the total weight of all bee combs at 23:00 every day, and the red dashed line represents the daily average of inside temperature of a bee hive. The humidity inside the hive is not shown because the curve is complicated due to multiple influence factors, such as ambient conditions and bee activities. According to the monitoring data, the bee

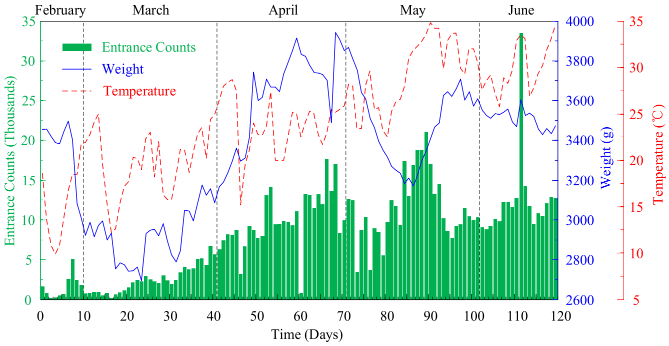

Fig. 9. Long-term monitoring data.

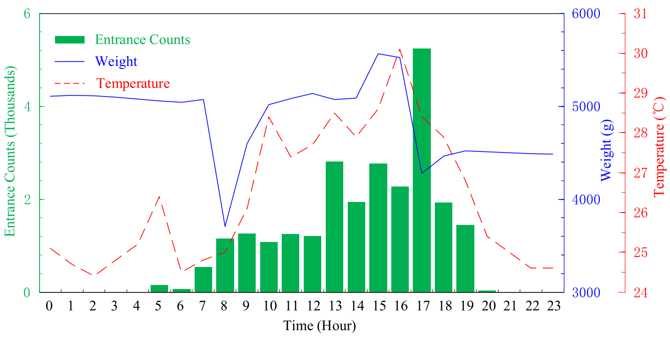

Fig. 10. Monitoring data for a swarming activity.

colony's activities can be divided into four stages. In the first 20 days, only a few bees went out for foraging every day because only a few flowers were booming in the early spring, meanwhile, a lot of food were consumed to feed the larvae and maintain internal temperature, which makes the weight of the bee combs rapidly went down in this period. During the 20th–60th days, the available food was becoming more abundant because the ambient temperature was rising at that time, which led to the increase of the entrance counts and combs' weight. During the 60th–100th days, the food outside became short again due to the drying up of nectar sources in that period, and the bee colony tried to balance the supply and the demand of food through optimizing foraging intensity and breeding scale. During the 100th–120th days, the entrance counts and weight were basically stable after the optimizing.

B. Swarming Activity Observation

Fig. 10 presents the monitoring data of the bee colony numbered 6048 on April 12, 2018, where the green bar represents the hourly entrance counts, the blue solid line represents the hourly weight of the total bee combs, and the red dashed line represents the hourly inside temperature of the bee hive. On this day, a successful swarming was observed by the beekeeper between 16:00 and 17:00, which was also evidently indicated by a series of synchronous variations in weight, entrance counts, and temperature. Meanwhile, almost the same variation of weight appeared between 7:00 and 8:00 without a large entrance counts, which means that some bees held honey had left their combs but still stayed in the hive for the command of swarming. The reason for the unsuccessful

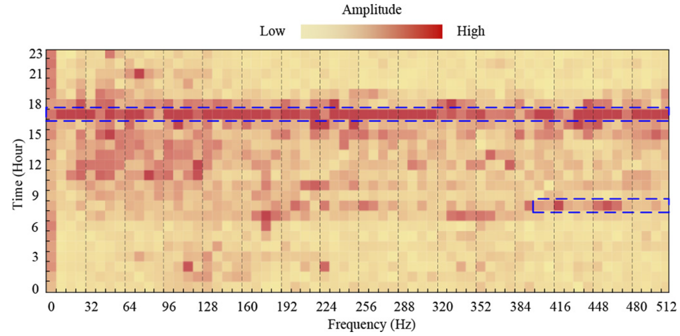

Fig. 11. Sound analysis for swarming activity.

swarming in the morning might be unsuitable ambient conditions or inconsistent decisions. After 10:00, the weight almost returned to the original value, which means that the bees were back to their routine work again after the unsuccessful swarming. As the weight reductions at 8:00 and 16:00, 1366g and 1248g, were quite similar, we speculate that the two actions may be launched by the same group of bees, i.e., the individual willing of the swarming may depend on a long-term feeling about the living conditions rather than the transient motivating at that moment.

Fig. 11 provides the spectrum analysis for the colony's sounds on that day, where the horizontal axis represents the frequency range of 0–512 Hz with the resolution of 8 Hz, the vertical axis represents the time of collecting data, and the color of each pixel represents the amplitude of sounds at each frequency resolution, i.e., the deeper of color means the louder of sound. Apparently, the most of deep pixels locate in the period between 7:00 and 19:00, which means the colony's noise during that period is louder due to the daily activities. Especially, the color between 16:00 and 17:00 is deeper than that of other times almost in the whole frequency range, in which the swarming was observed. The color in the frequencies over 400 Hz between 7:00 and 8:00 is also different from that of other intervals, which is also corresponding to the potential activities of swarming. Therefore, we propose the amplitude of the frequencies of 400–500 Hz as an indicator for swarming activity, which is consistent with Esch's study [30].

V. CONCLUSION

Based on the investigation of the connection between detectable features and bee colony's activities, this article presented an IoT-based system for long-term and extensive monitoring of bee colonies, which can hourly detect the temperature and humidity inside the hive, bee combs' weight, bee colony's sounds, and the counts of bees passing through the hive entrance. Compared with the reported monitoring system, this system is superior in the diversity of detected features, the capability of power self-support, and the wireless of data transmission that will benefit the system deployment in the field and long-term operation without maintenance.

The experiment of long-term monitoring has indicated that the system can continuously operate without human intervention. The temperature and humidity can reflect the microclimate of the bee hive, the weight can show the forging and stock of food, and the entrance counts are strongly related to the activeness and scale of bee colony, especially, to combining the sound of the bee colony, the bee colony's activities, such as swarming, can be effectively identified even predicted. Therefore, the proposed system has shown significant effects to confirm the activities and statuses of bee colonies, which could contribute to the study of bee's behavior and the development of precision beekeeping. In the future, more smart hives will be deployed in the field and machine learning will be applied to the monitoring data.

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